LAB 6: Sequence Diagram

Smart Waste Collection Management System

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1. Objective

To draw UML sequence diagrams for further scenarios of the Smart Waste Collection Management System, showing how the administrator interacts with the dashboard, backend server, database and sensor nodes over time, based on SRS v1.0.

2. Sequence diagrams included

Figure	Scenario	SRS reference
Diagram 1 (page 2)	Administrator logs in (with optional OTP), views the colour-coded bin map and generates a collections / average fill-level report. Invalid credentials are rejected.	4.4, 5.3 (REQ-DB-1 to 3)
Diagram 2 (page 3)	Administrator sets the alert threshold per bin or globally. Only authorized admins can change it, the change is logged, and it applies from the next sensor reading.	4.2, 5.5 (REQ-AL-1, REQ-AL-3)

3. Lifelines (participants)

Participant	Type	Role in the interaction
Administrator	Actor	Municipal administrator using the web dashboard.
Web Dashboard	Object	Browser interface: login, live bin map, reports and threshold settings.
Backend Server	Object	Authenticates users over HTTPS, checks the admin role, builds reports, updates thresholds.
Database	Object	Stores admin accounts, bins, collection records, thresholds and the change log.
Sensor Node	Object	Sends the next fill-level reading, which is checked against the new threshold.

4. Notation used

Symbol	Meaning
Dashed vertical line / box on top	Lifeline of a participant (green = system object, blue = actor)
Thin rectangle on a lifeline	Activation bar - the participant is executing
Solid arrow with filled head	Synchronous message (call)
Dashed arrow with open head	Return message
Arrow looping back to the same lifeline	Self message
alt / opt frames	Alternatives (if / else) and optional behaviour, with guard conditions

Sequence Diagram 1 - Administrator Login and Dashboard Analytics

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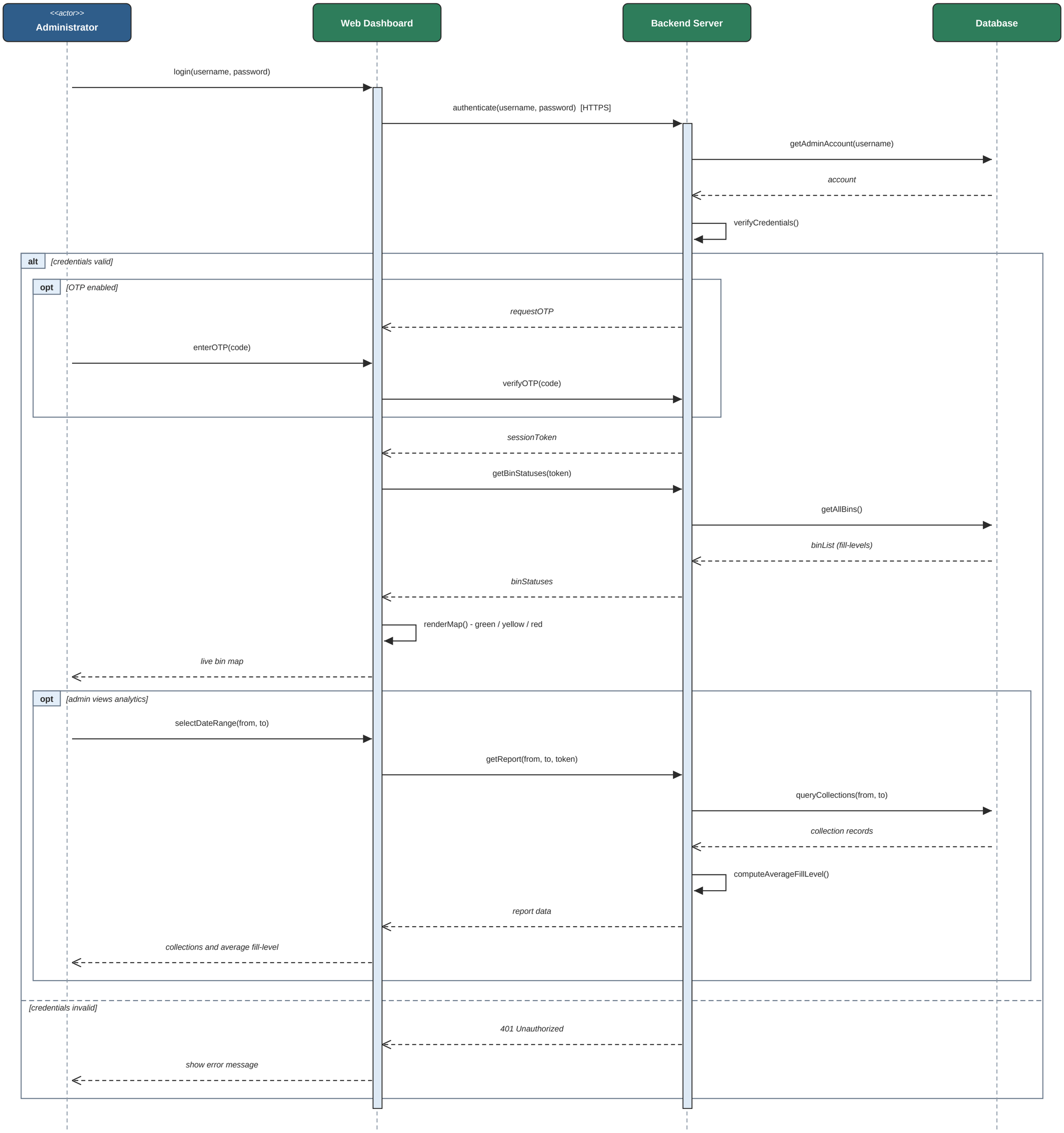

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Sequence Diagram 2 - Configuring the Alert Threshold

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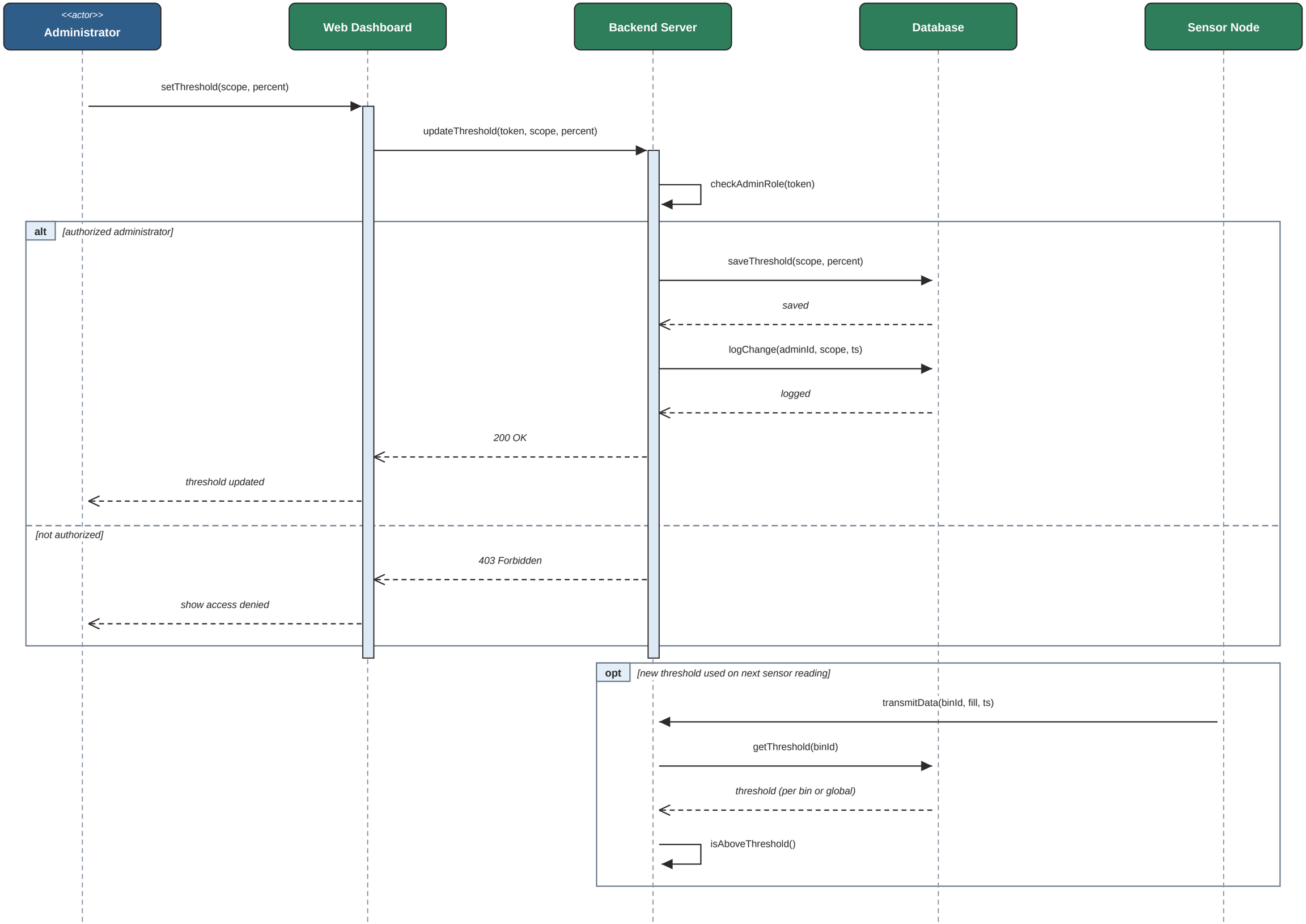

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